

JORDAN MONTENEGRO-CALLA

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I build clinical AI for the gap between high-resource medicine and the patients who never reach it. After two years of medical school at Universidad de San Martín de Porres in Peru, I moved to the United States and completed a B.S. in Medical Microbiology at Weber State University, then added formal training in software engineering and computer science to translate clinical reasoning into deployable systems. Today, I lead Kallpa, a bilingual Spanish-English primary care decision-support platform under development at kallpahealthcare.com, and I built Amoebanator, a calibrated binary triage classifier for primary amoebic meningoencephalitis risk that uses conformal prediction and selective abstention. I am extending this work to a cross-lingual study on real clinical data (MIMIC-IV), testing whether conformal coverage guarantees calibrated in English hold under distribution shift to Spanish. My research focus is clinical machine learning safety for bilingual and underserved populations, anchored in Peruvian and Latin American primary care.

RESEARCH EXPERIENCE

Independent Research | Amoebanator: Calibrated Triage Classifier for Primary Amoebic Meningoencephalitis (PAM) Risk

Role: Lead Researcher | 2025 to Present

- Designed a Pydantic v2 data schema for a literature-anchored meningitis/encephalitis vignette corpus, defining a 9-class label space (bacterial, viral, tuberculous, cryptococcal/fungal, granulomatous amebic encephalitis, neurocysticercosis, cerebral malaria, and non-infectious mimics), with primary amoebic meningoencephalitis (PAM) as the single class the current model targets.
- Implemented cross-field validators, including a CDC-PAM-2017 freshwater-exposure rule and a Cohen kappa inter-rater field, with primary amoebic meningoencephalitis treated as the highest-stakes target class.
- Implemented the classifier in PyTorch as a compact tabular multilayer perceptron (914 parameters) with temperature scaling for calibration.
- Implemented out-of-distribution detection (Mahalanobis distance and energy-based gates) and conformal prediction (split and label-conditional Mondrian), with a finite-sample regime badge (ASYMPTOTIC / FINITE-SAMPLE / INVALID) that abstains when the calibration set is too small for a valid coverage guarantee, as is the case at the current calibration size of $n=6$.
- Trained the model on a separate small synthetic tabular set ($n=30$, distinct from the vignette corpus, drawn from published case-series marginals in Yoder 2010, Cope 2016, and CDC data) with a train/validation split and zero real PHI. Maintained a test-driven, continuously integrated build. The conformal-prediction and selective-abstention methodology developed here is being carried forward into a follow-up study on real-world clinical data.
- GitHub: github.com/ljm234/amoebanator-25 | HuggingFace Space: huggingface.co/spaces/luisjordanmontenegro/amoebanator-25

SALUD: Cross-Lingual LLM Uncertainty and Deferral for Mental-Health Screening in Spanish and Quechua

Lead Researcher (Independent) | 2026 to Present

- Designing a cross-lingual study of large language model uncertainty quantification and selective deferral for depression and anxiety screening (PHQ-9 and GAD-7) across Spanish and Quechua, using clinical vignettes to reduce institutional review board burden.
- Leading an interdisciplinary team of two medical students and a native Quechua-speaking senior physician to develop culturally adapted screening vignettes.
- Grounded in a documented gap identified through literature and documentation review: no major large language model officially documents Quechua support, and no clinical Quechua screening benchmark currently exists.

Naylamp: Distributed Vector Database (Independent Systems Engineering Project)

Sole Developer | 2026 to Present

- Building a distributed vector database from scratch in Go with zero external dependencies (standard library only), spanning an HNSW approximate-nearest-neighbor index, a crash-safe write-ahead-log storage engine with atomic snapshots and recovery, and a from-scratch Raft consensus layer.
- Implemented the Raft consensus algorithm from first principles as a pure, side-effect-free core (no goroutines, timers, or wall-clock time in the state machine), covering leader election, log replication, durable state with a segmented CRC log, chunked snapshot installation, PreVote, and ReadIndex linearizable reads, enabling fully deterministic replay under simulation.
- Designed a deterministic simulation-testing harness that verified the four Raft safety invariants across 300 seeded runs under injected message loss, duplication, reordering, and network partitions, with a schedule-coverage gate to prevent illusory coverage.
- Engineered a crash-safe persistence layer (segmented write-ahead log with fsync durability, atomic snapshots with manifest, evidence-based compaction, and crash-injection recovery gates), verified by a test that kills and restarts the leader and confirms zero acknowledged writes are lost.
- Maintained rigorous engineering discipline across 35 GPG-signed, verified commits with a phased roadmap and a formal deferred-backlog protocol.

Computational and Experimental Investigation of ER Stress as a Therapeutic Vulnerability in *Naegleria fowleri*

Role: Lead Researcher | Grant: \$10,500 total | Weber State University, Ogden, UT | 2024 to 2025

- Screened multiple candidate pathways (RNA polymerase subunits, ribosomal RNAs, eIFs, Ran-GTPase, Sar1/COPII secretory machinery) to identify proteins where *N. fowleri* and human homologs diverge enough to support selective drug targeting, using sequence alignment and 3D structural comparison in NCBI iCn3D and PyMOL.
- Focused on the Leucyl-tRNA synthetase (LeuRS) editing domain (CP1) and mapped hydrophobic and electrostatic surfaces to define a parasite-specific pocket around key residues that shows 9 to 13 Å spatial divergence from the human enzyme.
- Developed a therapeutic hypothesis in which selective LeuRS inhibition triggers ER stress and collapse of the trophozoite, with practical delivery strategies including liposomal formulation and intrathecal/CSF delivery.
- Validated the ER stress mechanism experimentally using tunicamycin- and thapsigargin-induced ER stress models, plus metformin-driven mitochondrial dysfunction. Quantified stress responses by Western blot (BiP/GRP78) and JC-1 mitochondrial membrane potential staining.
- Manuscript in preparation. Preliminary findings deposited at Zenodo: <https://zenodo.org/records/17527970>

Lead Researcher | Montenegro's Medium: An Optimized, Low-Cost Culture Formulation for Axenic Growth of *Naegleria fowleri*

Weber State University, Ogden, UT | 2025

- Developed Montenegro's Medium, an optimized, low-cost culture formulation for axenic growth of *Naegleria fowleri*, reducing costs by over 70 percent compared to commercial alternatives.
- Conducted growth kinetics analysis, achieving approximately 3 times higher cell yield (1.03×10^7 cells/mL at 96 hours) than standard media formulations.
- Performed cell culture maintenance, viability testing, and morphological analysis using phase-contrast microscopy.
- Contributed to protocol optimization for filter sterilization and stock solution preparation to improve reproducibility.
- Submitted to Journal of Eukaryotic Microbiology. Open preprint: <https://doi.org/10.5281/zenodo.17766277>

***Naegleria fowleri* Drug Synergy Studies in a Rescue-of-Infection Model**

Research Assistant | Weber State University, Ogden, UT | 2023 to 2024

- Tested Amphotericin B and Miltefosine combinations in a HeLa rescue-of-infection model, achieving 80 percent host cell survival; co-authored abstract published at UCUR Symposium 2024.
- Methods: LDH cytotoxicity assays, fluorometric caspase-3 analysis.

Twing Lab | Polychlorinated Biphenyl Contamination Studies

Participant | Weber State University, Ogden, UT | 2023

- Conducted environmental contaminant surveillance in Weber County using EPA Method 8082A and QuEChERS extraction coupled with GC-MS analysis; quantified PCB-138/153/180 congener distributions in Woods Pond sediments, characterizing persistent weathering profiles decades after industrial release.

PRESENTATIONS & PUBLICATIONS

Manuscripts Under Review

Montenegro-Calla, L. J., & Jara-Scholz, J. A. (2025). Montenegro's Medium (MM): A Serum-Free, Low-Cost Axenic Medium for Robust Growth of *Naegleria fowleri*. *Journal of Eukaryotic Microbiology* (under review). Open preprint: <https://doi.org/10.5281/zenodo.17766277>

In Preparation

Montenegro-Calla, L. J. (2026). Amoebanator: A Calibrated Binary Triage Classifier for Primary Amoebic Meningoencephalitis Risk Using Conformal Prediction and Selective Abstention. medRxiv preprint targeted for 2026.

Montenegro-Calla, L. J. (in preparation). Computational and Experimental Investigation of ER Stress as a Therapeutic Vulnerability in *Naegleria fowleri*: From LeuRS Drug Targeting to Trophozoite Collapse. Preliminary findings: <https://doi.org/10.5281/zenodo.17527970>

Oral Presentations

Montenegro-Calla, L. J. (2026, April). An Integrated Pipeline for *Naegleria fowleri*: Computational Drug Target Discovery, Low-Cost Axenic Culture, Calibrated AI Triage, and Bilingual Clinical Decision Support. National Conference on Undergraduate Research (NCUR 2026), Richmond, VA, April 13 to 15, 2026. Selected by Weber State University with full institutional travel funding.

Montenegro-Calla, L. J. (2026, February). Low-Cost Axenic Culture of *Naegleria fowleri*, Computational Drug Targeting, and AI-Enabled Triage for Primary Care in Underserved Settings. Utah Conference on Undergraduate Research (UCUR 2026), Weber State University, February 27, 2026.

Montenegro-Calla, L. J. (2024). Brain-Eating Amoeba Rescue-of-Infection Model Using Drug Combinations. Utah Conference on Undergraduate Research (UCUR 2024). Co-author. Demonstrated combinatorial efficacy of Amphotericin B and Miltefosine in HeLa rescue-of-infection model.

Montenegro-Calla, L. J. (2023). Two presentations at Weber State University Symposium 2023: "Naegleria fowleri: Drug Inhibition in a Human Cell Infection Model" (Presenter, characterized Rifampin's dose-dependent efficacy in HeLa rescue-of-infection model); "PCB Contamination in Weber County: GC-MS Analysis" (Lead Presenter, interpreted GC-MS results across Weber County).

Montenegro-Calla, L. J. Presenter, Weber State University Symposium, Ogden, UT, 2024, 2025, and 2026.

EDUCATION

University of the People: B.S. Computer Science | Expected Oct 2026

GPA: 3.92. Coursework: analysis of algorithms, artificial intelligence, machine learning, data mining, databases, and cybersecurity. Frameworks: TensorFlow, PyTorch.

Ensign College, Salt Lake City, UT: B.S. Software Engineering | Expected Oct 2026

GPA: 4.000. Highest Honors. Coursework: discrete mathematics, probability and statistics, data structures and algorithms, software architecture, DevOps. Stack: Python, SQL, Docker, JavaScript/TypeScript, AWS/Azure.

Ensign College, Salt Lake City, UT: B.S. Information Technology and System Administration | December 2025

GPA: 4.000. Highest Honors. Coursework: cloud server administration (AWS, Azure), networking, IT architecture, cybersecurity operations.

Weber State University, Ogden, UT: B.S. Medical Microbiology (Pre-Med Track) | August 2025

GPA: 3.57. Honors: Wildcat Advantage Highest Level (Presidential Letter Recipient), Excellence in Community Engagement Award. Quantitative coursework: Calculus I. Lab techniques: CRISPR-Cas9, DNA sequencing, mass spectrometry, flow cytometry, confocal microscopy, Western blotting, PCR.

Universidad de San Martín de Porres, Chiclayo, Peru: Faculty of Human Medicine (M.D. Program) | 2016 to 2018

Completed two years of clinical medical education before transferring to the United States. Coursework: Cell and Molecular Biology, Human Anatomy I-II, Physiology I, Biochemistry, Medical Procedures, Public Health.

CERTIFICATIONS & CREDENTIALS

Research Database Access

PhysioNet MIMIC-IV Database Access (PhysioNet/MIT). Credentialed April 2026.

Specialized Coursework

Stanford University Online: AI in Healthcare Specialization (54.5 CME credits, 2025). Stanford Center for Health Education. Courses: Introduction to Healthcare; Introduction to Clinical Data; Fundamentals of AI and Machine Learning in Healthcare; Evaluations of AI Applications in Healthcare; AI in Healthcare Capstone Project.

DeepLearning.AI: Deep Learning Specialization (Andrew Ng, 2025). Courses: Neural Networks and Deep Learning; Improving Deep Neural Networks; Structuring Machine Learning Projects; Convolutional Neural Networks; Sequence Models.

DeepLearning.AI: AI for Medicine Specialization (Andrew Ng, 2025). Courses: AI for Medical Diagnosis; AI for Medical Prognosis; AI for Medical Treatment.

DeepLearning.AI: Machine Learning Specialization (Andrew Ng, 2025). Courses: Supervised Machine Learning; Advanced Learning Algorithms; Unsupervised Learning, Recommenders, and Reinforcement Learning.

Technical Certifications

TestOut Hybrid Server Pro: Core (2025); TestOut Linux Pro (2025).

AWARDS & HONORS

- Highest Honors, Ensign College (B.S. Information Technology and System Administration, GPA 4.000), 2025.
- Wildcat Advantage Highest Level, Presidential Letter Recipient, Weber State University, 2025.
- Excellence in Community Engagement Award, Weber State University, 2023.
- President's List, University of the People (B.S. Computer Science, GPA 3.92), 2025 to 2026.

Leadership Roles

- Community Leader Intern, Ogden Civic Action Network (OgdenCAN), Ogden, UT, 2022 to 2023. Led the "Take Your Time on Taylor" neighborhood revitalization project.

ENTREPRENEURSHIP & INNOVATION

Top 10 Finalist, Lion's Den Entrepreneur Challenge, Ensign College, Salt Lake City, UT, January 2025. Selected from 90 startup teams; pitched Kallpa Healthcare and was awarded prize funding to support prototyping, early pilots in low-resource clinical settings, and continued development.

CLINICAL EXPERIENCE

Medical Student, Community Health Clinics, Chiclayo, Peru | 2016 to 2018

- Shadowed family medicine physicians for over 250 hours across hospital, outpatient, and community health settings.
- Performed venous blood collection (phlebotomy) and vital signs assessment under physician supervision.
- Supported patient intake, clinical documentation, and health education in primary care delivery.

- Trained in basic life support, first aid, and airway management (positioning, bag-mask ventilation); observed intubation procedures under physician supervision.

RESEARCH INTERESTS

- Clinical machine learning, with focus on calibration, selective prediction with abstention, conformal prediction, out-of-distribution detection, and algorithmic fairness.
- Bilingual (Spanish-English) and multilingual clinical decision support for underserved populations across Latin America and the United States.
- Multimodal clinical machine learning combining clinical text, laboratory measurements, and structured electronic health record data.
- Human-AI collaboration and workflow integration in primary care, with a focus on low-resource and LMIC settings.

SKILLS

Technical Skills

- **Machine Learning:** PyTorch, deep learning (CNNs, RNNs, transformers), ensemble methods (random forest, gradient boosting), conformal prediction (split and Mondrian), calibration (temperature scaling, Platt, isotonic), out-of-distribution detection (Mahalanobis and energy-based), selective prediction with abstention, Bayesian inference.
- **Programming Languages:** Python, TypeScript, JavaScript, R, SQL, Bash.
- **Frameworks and Libraries:** Streamlit, FastAPI, Next.js, Vitest, Auth.js, Pydantic v2, Matplotlib, Plotly.
- **Data and Infrastructure:** PostgreSQL (Neon), MIMIC-IV (credentialed April 2026), multi-tenant database architectures, schema design and migration management, Hugging Face Spaces, GitHub Actions (CI/CD), Linux server administration.

Clinical and Research Skills

- **Pathophysiology and clinical reasoning:** infectious disease mechanisms, host-pathogen interactions, drug synergy, symptom pattern recognition, differential diagnosis development.
- **Laboratory:** cell culture, microbiological techniques, growth kinetics analysis, microscopy, environmental microbiology, protocol development.
- **Research methodology:** experimental design, statistical analysis, technical writing (preprints, conference abstracts), independent project development.

Languages

English (fluent), Spanish (native).

PROFESSIONAL EXPERIENCE

Founder & Lead Developer, Kallpa Healthcare: Bilingual Primary Care Decision Support

Role: Independent Project | 2025 to Present

- Built a bilingual Spanish/English clinical decision support platform for Peruvian primary care, deployed at kallpahealthcare.com, with Auth.js multi-tenant authentication (Argon2id + WebAuthn passkeys, row-level security, 43-table Drizzle schema with clinic-scoped isolation).
- Authored a 358-disease clinical knowledge base with 3,322 finding-level likelihood ratios anchored to 2,024 PMID citations (ICD-10, CIE-10, ICD-11), with citation validation that cross-references NCBI eSummary, Retraction Watch, and OpenAlex. This knowledge base is staged for the V1.5 diagnostic engine; the V1.0 release uses an LLM-based diagnostic surface with retrieval-augmented generation (OpenAI embeddings, pgvector).
- Built Peru-native clinical infrastructure: UBIGEO altitude-band engine (WMS 2024 dosing), endemic-disease dispatcher (bartonellosis, leishmaniasis, malaria, dengue, oropouche), CIE-10 auto-coder, MINSA NTS protocol RAG, DIGEMID PNUME formulary, and DS-016 compliance scaffolding.

- Designed a bias-audit framework (disparate impact and equal-opportunity TPR deltas across language and geography) and maintained Vitest test coverage. V1.0 launch planned for 2026.
- In development: kallpahealthcare.com |

Supplemental Instruction (SI) Leader and Curriculum Developer, MAT 107: Quantitative Analysis
Ensign College, Salt Lake City, UT | 2025

- Taught Quantitative Analysis to undergraduate students through weekly Supplemental Instruction sessions covering descriptive statistics, applied problem-solving, and quantitative reasoning.
- Co-developed the MAT 107 course curriculum with faculty: learning outcomes, weekly modules, formative and summative assessments, and final exam materials.
- Designed exam preparation strategies and study materials targeting common learning gaps for struggling students.

Canvas Accessibility Specialist

WSU Online, Ogden, UT | July 2023 to August 2025

- Reformatted over 1,500 scientific documents and lab manuals across STEM courses to meet WCAG 2.1 Level AA accessibility compliance, with primary focus on remediation.
- Collaborated with faculty in biology, chemistry, microbiology, and physics to improve course materials for students with visual, motor, and cognitive disabilities.
- Used Adobe Acrobat Pro and Microsoft Accessibility Checker for PDF remediation, reading-order correction, alternative-text generation, and compliance validation.

Propel Teacher Intern, High School STEM

WSU Propel Project, Ogden, UT | September 2022 to March 2023

- Taught and assisted with instruction in chemistry, mathematics, and microbiology for high school students alongside a lead teacher. Accumulated over 90 hours of combined classroom and laboratory teaching time.

VOLUNTEER SERVICE (600+ verified hours, GivePulse)

Full-Time Volunteer Missionary, Chile Osorno Mission

The Church of Jesus Christ of Latter-day Saints, Osorno, Chile | 2018 to 2020

- Lived and served full-time for two years in rural Chilean communities, working in Spanish with families across socioeconomic and health-access conditions; developed cross-cultural fluency and direct exposure to barriers low-resource communities face accessing basic health services.
- Taught community classes on health, family wellness, and education topics; led service projects supporting more than 600 families in underserved rural areas.

Food Security Volunteer, Weber Cares Food Pantry

Weber State University, Ogden, UT | 2023 to 2024

- Organized weekly food drives serving over 250 students experiencing campus food insecurity.
- Contributed to a measured 30% improvement in campus food security access during the program period.